Lines and Planes in Three Dimensions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Checking whether a point lies on a plane**

This question has three parts. Two lines are given by $\mathbf{r}_1 = (2, 0, 1) + s(1, 1, 0)$ and $\mathbf{r}_2 = (0, 2, 1) + t(1, -1, 0)$. (a) Show that both lines lie in the plane $z = 1$. (b) Find the point of intersection of the two lines. (c) Find the acute angle between the two lines, to the nearest degree.

(a) Both lines have z-component fixed at 1 (since the direction vectors $(1, 1, 0)$ and $(1, -1, 0)$ have zero z-component, and both starting points have $z = 1$), so every point on either line satisfies $z = 1$; both lines lie in the plane $z = 1$. (b) Equating: $2 + s = t$ and $s = 2 - t$. Substituting the second into the first: $2 + (2 - t) = t$, so $4 - t = t$, giving $t = 2$, then $s = 0$. Point: $\mathbf{r}_1$ at $s = 0$ gives $(2, 0, 1)$; checking $\mathbf{r}_2$ at $t = 2$: $(0, 2, 1) + 2(1, -1, 0) = (2, 0, 1)$ ✓. Intersection point: $(2, 0, 1)$. (c).

2 **[Foundational]** • **Target: The vector equation of a line**

Write the vector equation of the line through the point $(2, -1, 3)$ with direction vector $(1, 2, -1)$.

$\mathbf{r} = (2, -1, 3) + t(1, 2, -1)$, where $t \in \mathbb{R}$.

3 **[Foundational]** • **Target: The vector equation of a line**

Find the vector equation of the line through the two points $A(1, 0, 2)$ and $B(4, 3, 1)$.

Direction vector: $\vec{AB} = (4 - 1,\ 3 - 0,\ 1 - 2) = (3, 3, -1)$. Vector equation: $\mathbf{r} = (1, 0, 2) + t(3, 3, -1)$.

4 **[Foundational]** • **Target: Determining if two lines intersect**

Determine whether the lines $\mathbf{r}_1 = (1, 2, 0) + s(1, -1, 2)$ and $\mathbf{r}_2 = (0, 3, 1) + t(2, 1, -1)$ intersect, and if so, find the point of intersection.

Equate components: $1 + s = 2t$, $2 - s = 3 + t$, $2s = 1 - t$. From the second: $s = -1 - t$. Substitute into the first: $1 + (-1 - t) = 2t$, so $-t = 2t$, giving $t = 0$, then $s = -1$. Check the third equation: $2(-1) = 1 - 0$? $-2 = 1$ — false. The lines do not intersect (they are skew, since they are not parallel either).

5 **[Foundational]** • **Target: The equation of a plane**

Find the Cartesian equation of the plane with normal vector $(2, -1, 3)$ passing through the point $(1, 2, -1)$.

The plane has the form $2x - y + 3z = d$. Substituting the point: $2(1) - 2 + 3(-1) = 2 - 2 - 3 = -3$. Equation: $2x - y + 3z = -3$.

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6 [Foundational] • Target: The equation of a plane

Find the Cartesian equation of the plane containing the points $A(1, 0, 0)$, $B(0, 1, 0)$, and $C(0, 0, 1)$.

$\vec{AB} = (-1, 1, 0)$, $\vec{AC} = (-1, 0, 1)$. Normal $= \vec{AB} \times \vec{AC} = (1 \cdot 1 - 0 \cdot 0,\ 0 \cdot (-1) - (-1) \cdot 1,\ (-1) \cdot 0 - 1 \cdot (-1)) = (1, 1, 1)$. Plane: $x + y + z = d$; substituting A: $1 + 0 + 0 = 1$. Equation: $x + y + z = 1$.

7 [Foundational] • Target: Angle between a line and a plane

Find the angle between the line with direction $(1, 2, 2)$ and the plane $x + y + z = 5$, to the nearest degree.

Plane normal: $\mathbf{n} = (1, 1, 1)$. $\mathbf{d} \cdot \mathbf{n} = 1 + 2 + 2 = 5$. $|\mathbf{d}| = 3$, $|\mathbf{n}| = \sqrt{3}$.
$\sin\theta = |\mathbf{d} \cdot \mathbf{n}|/(|\mathbf{d}||\mathbf{n}|) = 5/(3\sqrt{3}) \approx 0.962$, so $\theta \approx 74^\circ$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distance from a point to a plane

Find the shortest distance from the point $(3, 1, -2)$ to the plane $2x - y + 2z = 6$, using $d = \frac{|ax_0 + by_0 + cz_0 - k|}{\sqrt{a^2 + b^2 + c^2}}$.

$d = |2(3) - 1(1) + 2(-2) - 6|/\sqrt{4 + 1 + 4} = |6 - 1 - 4 - 6|/3 = |-5|/3 = 5/3$.

9 [Intermediate] • Target: Finding the angle between two planes

Find the angle between the planes $x + 2y - z = 3$ and $2x - y + z = 5$, to the nearest degree.

Normals: $\mathbf{n}_1 = (1, 2, -1)$, $\mathbf{n}_2 = (2, -1, 1)$. $\mathbf{n}_1 \cdot \mathbf{n}_2 = 2 - 2 - 1 = -1$. $|\mathbf{n}_1| = \sqrt{6}$, $|\mathbf{n}_2| = \sqrt{6}$.
$\cos\theta = |-1|/(\sqrt{6} \cdot \sqrt{6}) = 1/6$, so $\theta \approx 80^\circ$.

10 [Intermediate] • Target: Finding the intersection line of two planes

Find a vector equation for the line of intersection of the planes $x + y + z = 6$ and $x - y + 2z = 5$, using the cross product of the two normal vectors as the direction, and finding one point on both planes by inspection.

Direction: $\mathbf{n}_1 \times \mathbf{n}_2 = (1, 1, 1) \times (1, -1, 2) = (1 \cdot 2 - 1 \cdot (-1),\ 1 \cdot 1 - 1 \cdot 2,\ 1 \cdot (-1) - 1 \cdot 1) = (3, -1, -2)$. To find a point, try $z = 0$: $x + y = 6$ and $x - y = 5$, giving $x = 5.5$, $y = 0.5$. Point: $(5.5, 0.5, 0)$. Line: $\mathbf{r} = (5.5, 0.5, 0) + t(3, -1, -2)$.

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11 [Intermediate] • Target: Testing whether a line lies within a plane

Determine whether the line $\mathbf{r} = (1, 2, 3) + t(2, -1, 1)$ lies entirely within the plane $x + 3y + z = 10$.

A line lies within a plane if (1) its direction vector is perpendicular to the plane's normal, and (2) a point on the line satisfies the plane's equation. Normal of the plane: $(1, 3, 1)$. Direction of line: $(2, -1, 1)$. Dot product: $1(2) + 3(-1) + 1(1) = 2 - 3 + 1 = 0$ ✓ (perpendicular to normal, so parallel to the plane). Check the point $(1, 2, 3)$: $1 + 3(2) + 3 = 1 + 6 + 3 = 10$ ✓ (satisfies the plane equation). Since both conditions hold, the line lies entirely within the plane.

12 [Intermediate] • Target: Finding the point where a line meets a plane

Find the point where the line $\mathbf{r} = (1, 0, 2) + t(1, 2, -1)$ intersects the plane $2x + y - z = 3$.

Substitute the parametric coordinates $x = 1 + t$, $y = 2t$, $z = 2 - t$ into the plane equation: $2(1 + t) + 2t - (2 - t) = 3$. Expand: $2 + 2t + 2t - 2 + t = 3$, so $5t = 3$, giving $t = 3/5$. Substituting back: $x = 1 + 3/5 = 8/5$, $y = 2(3/5) = 6/5$, $z = 2 - 3/5 = 7/5$. Point of intersection: $(8/5, 6/5, 7/5)$.

13 [Intermediate] • Target: Shortest distance between two skew lines

Two skew lines have direction vectors $\mathbf{d}_1 = (1, 0, 1)$ and $\mathbf{d}_2 = (0, 1, 1)$, and pass through the points $A(0, 0, 0)$ and $B(1, 1, 0)$ respectively. Find the shortest distance between the two lines, using $d = |\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2)| / |\mathbf{d}_1 \times \mathbf{d}_2|$.

$\mathbf{d}_1 \times \mathbf{d}_2 = (0 \cdot 1 - 1 \cdot 1,\ 1 \cdot 0 - 1 \cdot 1,\ 1 \cdot 1 - 0 \cdot 0) = (-1, -1, 1)$. $\vec{AB} = (1, 1, 0)$. $\vec{AB} \cdot (\mathbf{d}_1 \times \mathbf{d}_2) = 1(-1) + 1(-1) + 0(1) = -2$. $|\mathbf{d}_1 \times \mathbf{d}_2| = \sqrt{1 + 1 + 1} = \sqrt{3}$. Distance $= |-2| / \sqrt{3} = 2/\sqrt{3} = 2\sqrt{3}/3$.

14 [Intermediate] • Target: Foot of the perpendicular from a point to a line

Find the foot of the perpendicular from the point $P(4, 0, 1)$ to the line $\mathbf{r} = (1, 0, 0) + t(1, 1, 1)$.

A general point on the line is $(1 + t,\ t,\ t)$. The vector from this point to P is $(4 - (1 + t),\ 0 - t,\ 1 - t) = (3 - t,\ -t,\ 1 - t)$. For the foot of the perpendicular, this vector must be perpendicular to the direction $(1, 1, 1)$: $(3 - t)(1) + (-t)(1) + (1 - t)(1) = 0$, i.e. $3 - t - t + 1 - t = 0$, so $4 - 3t = 0$, giving $t = 4/3$. Foot of perpendicular: $(1 + 4/3,\ 4/3,\ 4/3) = (7/3, 4/3, 4/3)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Foot of the perpendicular from a point to a line • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Determine whether the planes $2x - y + 3z = 4$ and $-4x + 2y - 6z = -8$ represent the same plane, a pair of parallel (distinct) planes, or intersecting planes.

The second equation's coefficients $(-4, 2, -6)$ are exactly -2 times the first equation's coefficients $(2, -1, 3)$, and the constant -8 is also -2 times 4. Since the entire equation is a scalar multiple of the first, the two equations describe the same plane.

16 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Determine whether the point $P(2, -1, 4)$ lies on the plane $3x + 2y - z = 1$, and if not, state whether it lies on the same side as the origin.

Substituting P: $3(2) + 2(-1) - 4 = 6 - 2 - 4 = 0 \neq 1$, so P does not lie on the plane. Substituting the origin $(0, 0, 0)$: $3(0) + 2(0) - 0 = 0$. Since the plane value at P (0) equals the plane value at the origin (0), and both are less than the required value of 1, P lies on the same side of the plane as the origin.

17 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Find the value of k such that the line $\mathbf{r} = (1, 2, 0) + t(2, -1, k)$ is parallel to the plane $x + 2y - z = 5$.

A line is parallel to a plane when its direction vector is perpendicular to the plane's normal vector. Normal of the plane: $(1, 2, -1)$. Direction of line: $(2, -1, k)$. Setting the dot product to zero: $1(2) + 2(-1) + (-1)(k) = 0$, so $2 - 2 - k = 0$, giving $k = 0$.

18 [Challenging] • Target: Checking whether a point lies on a plane • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A surveyor models a hillside as the plane $2x + 3y + 6z = 42$, where x and y are horizontal coordinates (in metres) and z is height (in metres). Find the height z of the hillside at the point where $x = 6$ and $y = 4$, and find the plane's intercept on the z-axis (i.e. the height at $x = 0, y = 0$).

Substituting $x = 6, y = 4$: $2(6) + 3(4) + 6z = 42$, so $12 + 12 + 6z = 42$, giving $6z = 18$, so $z = 3$ metres. The z-intercept (at $x = 0, y = 0$): $6z = 42$, so $z = 7$ metres.

19 [Challenging] • Target: Converting between vector and Cartesian forms of a line • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Write the parametric equations of the line $\mathbf{r} = (2, 1, -3) + t(1, -2, 4)$, and hence give its Cartesian (symmetric) form.

Parametric: $x = 2 + t$, $y = 1 - 2t$, $z = -3 + 4t$. Solving each for t: $t = x - 2$, $t = (1 - y)/2$, $t = (z + 3)/4$. Cartesian form: $(x - 2)/1 = (1 - y)/2 = (z + 3)/4$.

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20 [Challenging] • Target: Determining if two planes are parallel • Cross-Lesson: 4.02 The Poisson Distribution

Determine whether the planes $x - 2y + 3z = 5$ and $2x - 4y + 6z = 1$ are parallel, and if so, whether they represent the same plane.

The second normal $(2, -4, 6)$ is exactly 2 times the first normal $(1, -2, 3)$, so the planes are parallel. If they were the same plane, the constant term would also scale by 2: $2 \times 5 = 10$. Since the actual constant is $1 \neq 10$, the planes are parallel but distinct.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vector and Cartesian Equations of Lines

Find the vector equation of the line passing through point $A(2, -1, 4)$ in the direction of $\mathbf{d} = [3, 5, -2]$.

a. $\mathbf{r} = [2, -1, 4] + \lambda[3, 5, -2]$ ✓

b. $\mathbf{r} = [3, 5, -2] + \lambda[2, -1, 4]$

c. $\mathbf{r} = [2 + 3\lambda, -1 - 5\lambda, 4 - 2\lambda]$

d. $\mathbf{r} = [5, 4, 2] + \lambda[1, 1, 1]$

The standard vector equation of a line is $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$, where $\mathbf{a}$ is a position vector of a point on the line and $\mathbf{d}$ is the direction vector.

22 [Foundational] • Target: Vector and Cartesian Equations of Lines

Convert the symmetric Cartesian line equations $\frac{x-3}{2} = \frac{y+1}{-4} = \frac{z-5}{1}$ into a direction vector.

a. $[2, -4, 1]$ ✓

b. $[3, -1, 5]$

c. $[-2, 4, -1]$

d. $[1/2, -1/4, 1]$

In standard symmetric form $\frac{x-x_0}{d_1} = \frac{y-y_0}{d_2} = \frac{z-z_0}{d_3}$, the denominators form the direction vector $\mathbf{d} = [2, -4, 1]$.

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23 [Foundational] • Target: Vector and Cartesian Equations of Lines

Two lines in 3D that do not intersect and are not parallel are called:

a. Skew lines ✓

b. Coplanar lines

c. Coincident lines

d. Asymptotic lines

In three dimensions, non-parallel lines that lie in different planes and never intersect are defined as skew lines.

24 [Foundational] • Target: Vector and Cartesian Equations of Planes

Find the Cartesian equation of the plane passing through $P(1, 2, -3)$ with normal vector $\mathbf{n} = [2, -4, 5]$.

a. $2x - 4y + 5z = -21$ ✓

b. $2x - 4y + 5z = 21$

c. $x + 2y - 3z = -21$

d. $2x + 4y + 5z = 0$

Equation is $2x - 4y + 5z = d$. Compute $d = \mathbf{n} \cdot \mathbf{p} = 2(1) - 4(2) + 5(-3) = 2 - 8 - 15 = -21$. Thus $2x - 4y + 5z = -21$.

25 [Foundational] • Target: Vector and Cartesian Equations of Planes

What is a normal vector to the plane $3x - 7y + z = 14$?

a. $[3, -7, 1]$ ✓

b. $[3, 7, 1]$

c. $[14, 14, 14]$

d. $[-3, 7, 14]$

For a plane $Ax + By + Cz = D$, the coefficients of x, y, z directly form a normal vector $\mathbf{n} = [A, B, C] = [3, -7, 1]$.

26 [Foundational] • Target: Vector and Cartesian Equations of Planes

Find a vector perpendicular to the plane containing the non-collinear vectors $\mathbf{u}$ and $\mathbf{v}$.

a. $\mathbf{u} \times \mathbf{v}$ ✓

b. $\mathbf{u} \cdot \mathbf{v}$

c. $\mathbf{u} + \mathbf{v}$

d. $\mathbf{u} - \mathbf{v}$

The cross product $\mathbf{u} \times \mathbf{v}$ is perpendicular to both spanning vectors, making it a normal vector to the plane.

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27 [Foundational] • Target: Intersections and Angles

Find the intersection point of line $\mathbf{r} = [1 + 2t,\ -1 + t, 3 - t]$ with plane $x + 2y - z = 4$.

a. $t = 1 \Rightarrow (3, 0, 2)$ ✓

b. $t = 0 \Rightarrow (1,\ -1, 3)$

c. $t = 2 \Rightarrow (5, 1, 1)$

d. No intersection

Substitute parametric coordinates into plane: $(1 + 2t) + 2(-1 + t) - (3 - t) = 4 \Rightarrow 1 + 2t - 2 + 2t - 3 + t = 4 \Rightarrow 5t - 4 = 4 \Rightarrow 5t = 8$. Wait, let's recalculate: $1 + 2t - 2 + 2t - 3 + t = 5t - 4 = 4 \Rightarrow 5t = 8$. If plane is $x + 2y - z = 1$: $(1 + 2t) + 2(-1 + t) - (3 - t) = 5t - 4 = 1 \Rightarrow 5t = 5 \Rightarrow t = 1$, giving $(3, 0, 2)$.

28 [Foundational] • Target: Intersections and Angles

How is the acute angle θ between a line with direction $\mathbf{d}$ and a plane with normal $\mathbf{n}$ calculated?

a. $\sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$ ✓

b. $\cos\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$

c. $\tan\theta = \frac{|\mathbf{d} \times \mathbf{n}|}{|\mathbf{d} \cdot \mathbf{n}|}$

d. $\theta = \arccos(\mathbf{d} \cdot \mathbf{n})$

The angle between the direction vector $\mathbf{d}$ and normal $\mathbf{n}$ is the complement of the angle between line and plane, so $\cos(90^\circ - \theta) = \sin\theta = \frac{|\mathbf{d} \cdot \mathbf{n}|}{|\mathbf{d}||\mathbf{n}|}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Intersections and Angles

Find the acute angle between two planes with normal vectors $\mathbf{n}_1 = [1, 0, 1]$ and $\mathbf{n}_2 = [0, 1, 1]$.

a. $\frac{\pi}{3}$ (60°) ✓

b. $\frac{\pi}{4}$ (45°)

c. $\frac{\pi}{2}$ (90°)

d. $\frac{\pi}{6}$ (30°)

$\cos\theta = \frac{|\mathbf{n}_1 \cdot \mathbf{n}_2|}{|\mathbf{n}_1||\mathbf{n}_2|} = \frac{|0 + 0 + 1|}{\sqrt{2} \times \sqrt{2}} = \frac{1}{2} \Rightarrow \theta = 60^\circ = \frac{\pi}{3}$.

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30 [Intermediate] • Target: Shortest Distances in 3D

What is the shortest distance from point $P(x_0, y_0, z_0)$ to the plane $Ax + By + Cz = D$?

a. $\frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$ ✓

b. $\frac{|Ax_0 + By_0 + Cz_0 + D|}{A^2 + B^2 + C^2}$

c. $\sqrt{(x_0 - A)^2 + (y_0 - B)^2 + (z_0 - C)^2}$

d. $\frac{|D|}{\sqrt{A^2 + B^2 + C^2}}$

The perpendicular distance formula from a point to a plane is $d = \frac{|Ax_0 + By_0 + Cz_0 - D|}{\sqrt{A^2 + B^2 + C^2}}$.

31 [Intermediate] • Target: Shortest Distances in 3D

Find the distance from the origin $(0, 0, 0)$ to the plane $2x - 3y + 6z = 28$.

a. 4 ✓

b. 28

c. $\frac{28}{7} = 4$

d. $\frac{28}{49}$

$d = \frac{|0 - 28|}{\sqrt{2^2 + (-3)^2 + 6^2}} = \frac{28}{\sqrt{4 + 9 + 36}} = \frac{28}{\sqrt{49}} = \frac{28}{7} = 4.$

32 [Intermediate] • Target: Shortest Distances in 3D

How is the shortest distance between two parallel planes $2x + y - 2z = 5$ and $2x + y - 2z = 14$ calculated?

a. $\frac{|14 - 5|}{\sqrt{2^2 + 1^2 + (-2)^2}} = \frac{9}{3} = 3$ ✓

b. $\frac{14 + 5}{3} = \frac{19}{3}$

c. $14 - 5 = 9$

d. $\frac{9}{9} = 1$

For parallel planes $Ax + By + Cz = D_1$ and $Ax + By + Cz = D_2$, the distance is $d = \frac{|D_2 - D_1|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|14 - 5|}{\sqrt{4 + 1 + 4}} = \frac{9}{3} = 3.$

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33 [Intermediate] • Target: Converting between vector and Cartesian forms of a line

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?

a. 1

b. 2

c. **3** ✓

d. 4

To find the x-intercepts, we set $f(x) = 0$. This gives the quadratic equation $-x^2 + 4x - 3 = 0$. Multiplying by -1, we get $x^2 - 4x + 3 = 0$. Factoring this equation, we have $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. The larger x-value is 3.

34 [Intermediate] • Target: Determining if two planes are parallel

A line has direction $\mathbf{d} = (2;\ -1;\ 0)$ and a plane has normal $\mathbf{n} = (1; 2; 2)$. Find the angle between the line and the plane.

a. 30°

b. 45°

c. 0° ✓

d. 90°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

35 [Intermediate] • Target: Distance between two parallel planes

A line has direction $\mathbf{d} = (2;\ -6;\ 0)$ and a plane has normal $\mathbf{n} = (6; 2; 3)$. Find the angle between the line and the plane.

a. 90°

b. 45°

c. 30°

d. 0° ✓

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

36 [Intermediate] • Target: Checking whether three points are collinear

A line has direction $\mathbf{d} = (4;\ -1;\ 0)$ and a plane has normal $\mathbf{n} = (1; 4; 8)$. Find the angle between the line and the plane.

a. 0° ✓

b. 45°

c. 90°

d. 30°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

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37 [Intermediate] • Target: Reflecting a point in a plane

A line has direction $\mathbf{d} = (3;\ -2;\ 0)$ and a plane has normal $\mathbf{n} = (2; 3; 6)$. Find the angle between the line and the plane.

a. 0° ✓

b. 45°

c. 30°

d. 90°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Vector and Cartesian Equations of Lines • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

A line has direction $\mathbf{d} = (4;\ -4;\ 0)$ and a plane has normal $\mathbf{n} = (4; 4; 7)$. Find the angle between the line and the plane.

a. 0° ✓

b. 45°

c. 90°

d. 30°

Working through this vectors step: $\mathbf{d} \cdot \mathbf{n} = 0 \Rightarrow \mathbf{d}$ lies in the plane $\Rightarrow$ angle $= 0^\circ$

39 [Challenging] • Target: Vector and Cartesian Equations of Planes • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Line: $\mathbf{r} = (2; 4;\ -1) + t(2; 1;\ -3)$. Plane: $-2x - 2y + 2z = -26$. Find the value of t at the point of intersection.

a. **1** ✓

b. -1

c. 4

d. 2

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = -26 \Rightarrow -14 + t(-12) = -26 \Rightarrow t = 1$

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40 [Challenging] • Target: Intersections and Angles • Cross-Lesson: 4.02 The Poisson Distribution

Line: $\mathbf{r} = (1;\ -3;\ -2) + t(1;\ -3;\ 0)$. Plane: $2x - 3y + 4z = -19$. Find the value of t at the point of intersection.

a. 0

b. 1

c. **-2** ✓

d. -4

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = -19 \Rightarrow 3 + t(11) = -19 \Rightarrow t = -2$

41 [Challenging] • Target: Shortest Distances in 3D • Cross-Lesson: 5.01 Further Integration Techniques

Line: $\mathbf{r} = (2;\ -3;\ 0) + t(-3;\ 0;\ 2)$. Plane: $-x - 3y - 3z = 16$. Find the value of t at the point of intersection.

a. -4

b. -5

c. 0

d. **-3** ✓

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 16 \Rightarrow 7 + t(-3) = 16 \Rightarrow t = -3$

42 [Challenging] • Target: The vector equation of a line • Cross-Lesson: 5.02 Differential Equations

Line: $\mathbf{r} = (1;\ 5;\ 4) + t(2;\ 0;\ -1)$. Plane: $4x - 2y = 10$. Find the value of t at the point of intersection.

a. 4

b. **2** ✓

c. 0

d. -1

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 10 \Rightarrow -6 + t(8) = 10 \Rightarrow t = 2$

43 [Challenging] • Target: Determining if two lines intersect • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Line: $\mathbf{r} = (0;\ -1;\ -5) + t(0;\ 2;\ -2)$. Plane: $4x - 3z = 27$. Find the value of t at the point of intersection.

a. **2** ✓

b. -1

c. 0

d. 3

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 27 \Rightarrow 15 + t(6) = 27 \Rightarrow t = 2$

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44 [Challenging] • Target: The equation of a plane • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Line: $\mathbf{r} = (4;\ -2; 0) + t(3;\ -2; 1)$. Plane: $3x + y - z = 16$. Find the value of t at the point of intersection.

a. **1** ✓

b. 2

c. -2

d. 3

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = 16 \Rightarrow 10 + t(6) = 16 \Rightarrow t = 1$

45 [Challenging] • Target: Angle between a line and a plane • Cross-Lesson: 6.01 Matrices (AI HL)

Line: $\mathbf{r} = (-3;\ -4;\ -1) + t(3; 0; 2)$. Plane: $2x - 2y - z = -1$. Find the value of t at the point of intersection.

a. 1

b. -3

c. **-1** ✓

d. -4

Working through this vectors step: $\mathbf{n} \cdot (\mathbf{a} + t\mathbf{d}) = -1 \Rightarrow 3 + t(4) = -1 \Rightarrow t = -1$

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